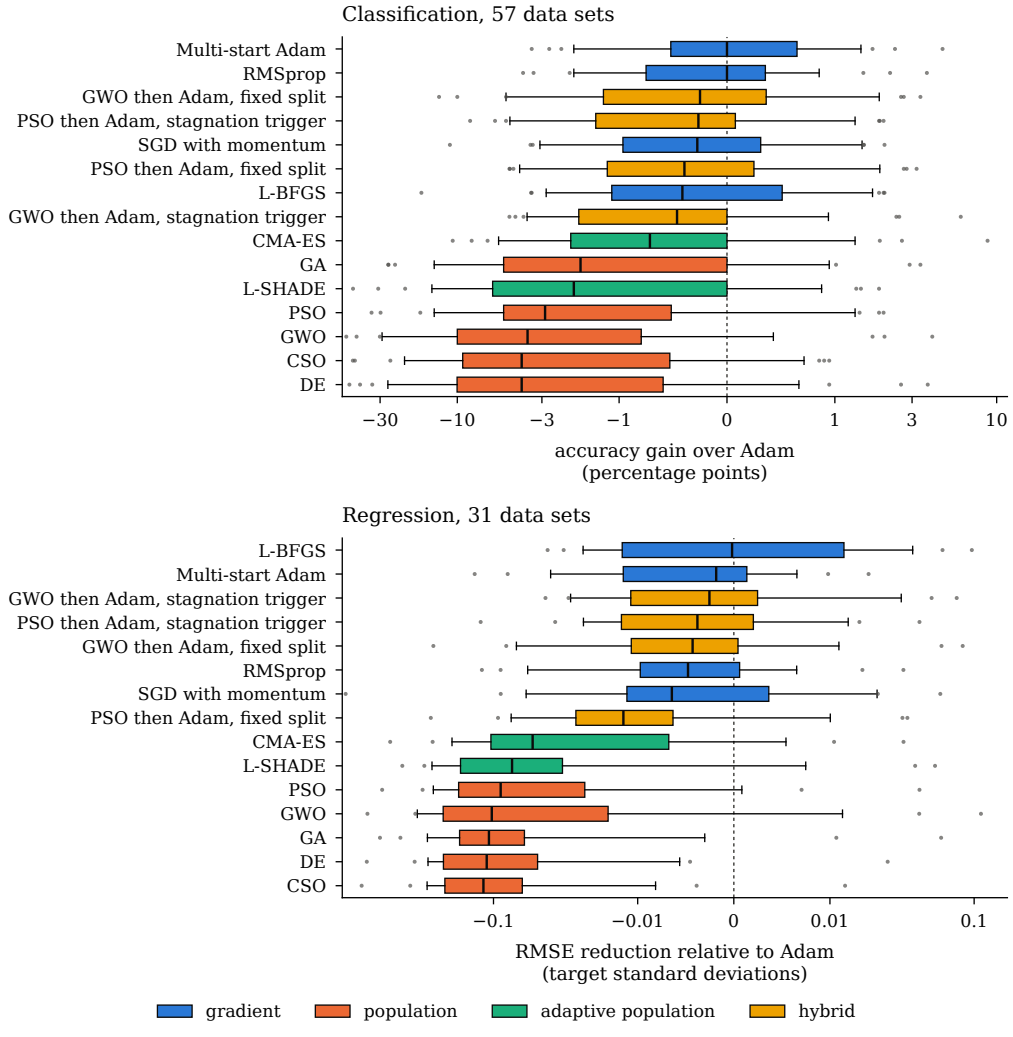
Classification, 57 data sets
Multi-start Adam
RMSprop
GWO then Adam, fixed split
PSO then Adam, stagnation trigger
SGD with momentum
PSO then Adam, fixed split
L-BFGS
GWO then Adam, stagnation trigger
CMA-ES
GA
L-SHADE
PSO
GWO
CSO
DE
−30
−10
−3
−1
0
1
3
10
accuracy gain over Adam
(percentage points)
Regression, 31 data sets
L-BFGS
Multi-start Adam
GWO then Adam, stagnation trigger
PSO then Adam, stagnation trigger
GWO then Adam, fixed split
RMSprop
SGD with momentum
PSO then Adam, fixed split
CMA-ES
L-SHADE
PSO
GWO
GA
DE
CSO
−0.1
−0.01
0
0.01
0.1
RMSE reduction relative to Adam
(target standard deviations)
gradient
population
adaptive population
hybrid